



Response to LlamaRisk on Discount-Based Oracle for Pendle PTs

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Note that the following assumes the oracle is only used for the AMM of Curve lending markets.

How the oracle choice affects Llamalend

We believe that the Oracle Choice is fine in certain scenarios.

However, we believe that there are issues when the **market price is significantly above the oracle price** as shown in your example:

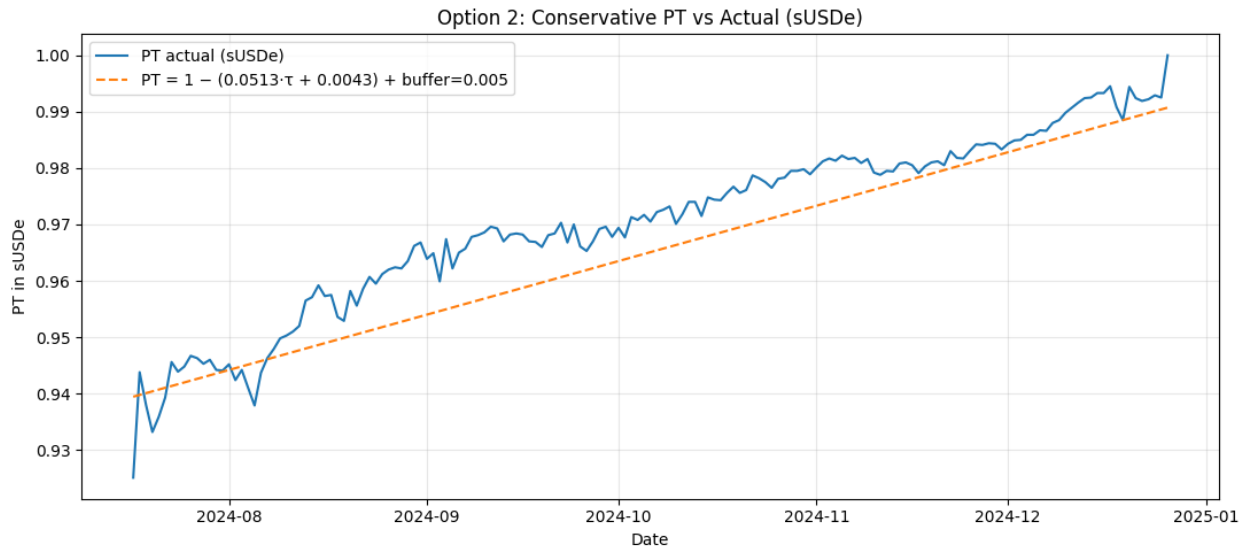


Figure 1: Source: <https://outline.llamarisk.com/s/59c9b021-6dc4-455b-910f-e77edc4ff5c6>

We believe that such situations in combination with the fact that the oracle price will continue to increase leads to **losses in Llamalend**.

We explain two scenarios that could cause losses in Llamalend. Importantly, we omit fees in these scenarios. We are aware that fees are important to consider and will help, but we don't think that they will eliminate the problem.

Scenario 1: Simple Arbitrage

Starting Situation:

- AMM Oracle Price: 0.93
- Market Price: 0.95
- Active Band is at 0.93
- Only collateral in AMM / no positions underwater

Arbitrageur buys PT at Llamalend AMM:

- Trades to collateral price of 0.925
 - Here they trade price is $p_o^{**3} / p_{up}^{**2} = 0.93^{**3} / 0.925^{**2} \sim 0.94$
- They sell the PT in the market for 0.95

Time passes and the Oracle Price rises while the Market price remains:

- AMM Oracle Price: **0.95**
- Market Price: 0.95
- Active Band is still at 0.925
- Some positions are partially liquidated due to previous step

Arbitrageur sells PT to Llamalend AMM:

- Buys PT at market price of 0.95
- Sells PT to Llamalend AMM at collateral price of 0.925
 - Here they trade for $p_o^{**3} / p_{up}^{**2} = 0.95^{**3} / 0.925^{**2} \sim 1.002$
 - **Llamalend might buy collateral for >1** which is never a good deal

Overall:

- LlamaLend has sold collateral for a price of 0.94
- Bought back that collateral for a price of 1.002
- Hence, the Llamalend borrowers sustain losses

Generally, we believe this will (to some degree) happen, whenever the market price is above the oracle price, because the arbitrageur moves the active band to a lower collateral price. This in combination with the rising oracle price will result in Llamalend buying back the collateral for a higher price.

Scenario 2: Buy and Hold

This scenario applies if the other market is very small or if no arbitraging is taking place between other market and Llamalend AMM.

If these conditions are fulfilled, someone who has studied the system can:

1. Buy collateral from Llamalend and hold it
2. Wait for the oracle price to go up
3. Sell the collateral back for a higher price

This works, because the trading price will increase due to the increasing oracle price. However, as mentioned, step 3 could be front-run by an arbitrageur.

Overall, we believe that a combination of:

- Predictably rising oracle price

and

- Higher market price with active arbitrage

or

- No arbitrage action

will generally be bad for Llamalend borrowers. Furthermore, we want to point out that Llamalend might end up buying PT for a price of more than 1 which is never a good deal for borrowers.

Oracle jumps

Regarding oracle jumps from parameter updates: bringing the oracle closer to the true market price is generally good, since it incentivizes arbitrageurs to (de)liquidate in line with the market. The real risk is when the update is so large that it creates a needless jump, and, smaller and gradual adjustments would have been enough to let arbitrage clear the gap without an outsized shock.

We think updates should be sized so any jump stays small enough that it doesn't create unnecessary arbitrage windfalls. The implemented AMM mitigations below are meant to absorb these jumps and further limit exploitable opportunities.

`limit_p_o` guards abrupt oracle moves by capping the applied jump and adding a short-lived “surge fee” that decays over ~120s.

Cap on immediate move

- $\text{ratio} = \min(p, \text{old_p_o}) / \max(p, \text{old_p_o})$
- If $\text{ratio} < 1/1.25$ (~20% drop or 25% rise), p_{new} is clipped to at most a $\sim \pm 25\%$ move and ratio is floored at ~ 0.8
- Prevents a single block from forcing an extreme price into the AMM

Dynamic (surge) fee

$\text{fee_dyn} = ((\text{old_dfee} + (1 - \text{ratio}^3)) * dt / \text{PREV_P_O_DELAY})$, capped just below 1

- $1 - \text{ratio}^3$ is the minimum fee to offset the jump (e.g., 20% drop $\rightarrow \sim 48.8\%$ fee)
- `old_dfee` carries prior surge fee forward if jumps pile up before decay finishes
- $dt / \text{PREV_P_O_DELAY}$ linearly decays the fee to zero over ~120s

Traders pay $\max(\text{base_fee}, \text{fee_dyn})$: right after a capped jump, arbitrage is expensive; if prices stabilize, the fee relaxes

Regarding oracle jumps, the usual trade-off applies where a higher fee can block undesired arbitrage but might also block intended soft liquidations.

Overall Conclusion

For both issues, given how parameter dependent the impact can be, we would recommend simulating a lending market using such an oracle together with active arbitrageurs to observe to which extent these issues can be problematic. In such a simulation different price variations and movements can be tested.

Best regards,
ChainSecurity Team